



GE Medical Systems

Technical Publications

Direction 2260859

Revision 2

Indigo to Octane Computer Upgrade

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Operating Documentation

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "**damage in shipment**" written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

Immediately complete "**Damage Loss Claim Form**" after damage is found. Form is available via MS Exchange Mail at Outlook/Public Folder/All Public Folders/Medical Systems/!Global Initiatives/Information Management/Forms/Common Forms/Damage Loss Claim Form. Send completed form to e-mail address listed on form.

For more information about the Transportation Claim Procedure, access the GE Medical Systems Intranet and enter the following URL address:

<ftp://3.87.40.2/globepro/qualsys/Docs/190016ME.PDF>

Rev. 11/15/2000



GE Medical Systems

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WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

- ・このサービスマニュアルには英語版しかありません。
- ・GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- ・このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
- ・この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意：

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

<u>REV</u>	<u>DATE</u>	<u>PRIMARY REASON FOR CHANGE</u>
0	Mar 4, 1999	Initial release.
1	Mar 7, 2002	Section 1 – Revised Octane parts lists. Section 3 – Revised software load instructions. Section 4 – Revised cable map by eliminating installation of Runs 792 and 809 (SPR's MRlge71765 & 72593). Section 5 – Removed references to Runs 792 and 809, updated ethernet cable connection information, and other minor changes.
2	Jan 15, 2003	Section 1 – Updated the EQUIPMENT RETURN PROCESS to indicate GoldSeal as the main source of contact. Updated the PRODUCT LOCATOR instructions. Added mobile information on ordering Octane mounting hardware from van manufacturers in UPGRADE SCHEDULE CONSIDERATIONS. Section 5 Changed return sheet title to GEMS GoldSeal.

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Direction 2128126	0*								
A	2								
I	0								
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1–4 to 1–5	1								
2–1	0								
3–1	1								
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* This revision number/letter corresponds to the indicated document's revision control system.

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SECTION 1 – UPGRADE INFORMATION

1-1 INTRODUCTION



WARNING!

FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS UPGRADE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

The Upgrade configuration documented in this publication is: Catalog M1033K, *Indigo to Octane Computer Upgrade*

<u>CURRENT SYSTEM CONFIGURATION</u>	➤	<u>UPGRADED SYSTEM CONFIGURATION</u>
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Signa Horizon LX with Indigo Computer	➤	Signa Horizon LX with Octane Computer
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1-2 EQUIPMENT RETURN PROCESS

The equipment or material being removed by the Octane Computer upgrade have a potential value for use in GoldSeal systems and service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

1. For disposition, contact:
 - GoldSeal – Contact the MR Product Specialist at GoldSeal Logistics phone number 877–251–4468. If GoldSeal is accepting the returned equipment then all transportation arrangements and packaging will be processed through them.
2. When calling, the following information is needed.
 - FDO (Future Delivery Order) number under which the upgrade was shipped.
 - Site name, address, and phone numbers.
 - Date of completed upgrade.
 - Name and Beeper number of Field Engineer that could be available when the equipment is picked up.
3. Regulations governing the disposal of this equipment or material vary from location to location and from country to country. In order to insure compliance with all local, state, or national environment, health, and safety regulations, dispose of all items in the per local GEMS policy and procedures and other applicable published guidance. Contact GEMS Recycling Center for assistance.
4. Process product locator information all serialized components removed from the system according to policy.

1-3 SITE READY CHECK

Pre-installation work must be completed before equipment is delivered to avoid delays and confusion. Refer to *Direction 2223751, Signa MR/i & CV/i Upgrade Pre-installation*, Pre-installation Checklist tab.

Tools and test equipment to install and calibrate the Signa Horizon 8.x system are listed in *Direction 2223751, Signa MR/i & CV/i Upgrade Pre-installation*, Section 10, TOOLS AND TEST EQUIPMENT.

Service installation tools used in this manual that may not be included in above references lists include:

- 46-301450G1 Fiber Optic Cable Connector Repair kit

1-4 PRODUCT DELIVERY INSTRUCTIONS AND PACKING

The "Shipping Document" lists catalog numbers delivered. **Review and confirm that order is delivered complete.** Check impact on installation schedule if Catalogs and/or packing boxes are missing and/or noted as shipped short.

Labels that are attached to the outside of packing boxes summarize box contents. The labels are color coded for different installation areas as follows:

- Green – Equipment Room
- Purple – Operator's Room
- Orange – Magnet Room
- Yellow – Post-installation (Phantoms, Coils).

"Product Delivery Instructions" (PDI) specify box contents, part numbers, and shipping procedure. The PDI is numbered according to the catalog number (for example, PDI – M1033K is for the Indigo to Octane Computer Upgrade). Lists of items included with each box are detailed by separate checklists, or a separate sheet that provides a summary of that box contents. Refer to PDI and packing lists for information specific to your shipment.

1-5 PRODUCT LOCATOR

The Global Install Base Database (also known as Product Locator System) tracks shipment, trans-shipment, and field location of the serialized models. There are now two methods for submitting Product Locator information.

1. At this time, for **U.S. ONLY**, the preferred method of submitting information is the *FE Site Verification Web Site* at: <http://gein2.med.ge.com/gib>

The FE Site Verification consists of three components that are available on the web from the main menu. They are:

- Install/deinstall product locator model and serial numbers
- Add/modify ship to address information
- Update CARES FE data for primary/secondary FEs

To obtain a copy of the FE training tutorial for using this website, a downloadable copy is available at the following: [Product Locator Support Central Page](http://supportcentral.ge.com/15563) – <http://supportcentral.ge.com/15563>

2. One "Shipping Card" is filled out and submitted when shipped (extra cards are supplied for trans-shipments between storage and distribution points), and the "Installation Card" and extra shipment cards are attached.

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

1-6 UPGRADE SCHEDULE CONSIDERATIONS

FMI's, periodic maintenance, installation of prerequisite upgrades and/or options (Multi-coil, Spectroscopy, etc.) will directly add to the man-hour and system down-time total. Be sure that all planned activity is considered when predicting the return of system to customer. Also verify Application Specialist is scheduled at the end of the installation to instruct users. Additional time must also be scheduled to instruct users following completion of upgrade.

Upgrade to Mobile systems require additional octane mounting hardware to be ordered from the van manufacturer. This hardware is required to protect the computer from the rigors of the mobile environment. To avoid delays, order the parts as soon as possible from either of the following:

- For vans built by Calumet Coach or AK Specialty Vehicles, order part number Octane Kit by contacting the Service department at 708-596-5066.
- For Ellis & Watts vans, call 513-752-9000 and order kit 91C-OPT-11.

1-7 UPGRADE SEQUENCE

Follow the Horizon system Upgrade Flow chart in SECTION 3 for best installation efficiency.

Note that many procedures may be performed in parallel and may be performed in any order according to the specific situation of each site. However all parallel tasks are to be completed prior to the next task.

Alternate tasks are noted above the task block to set them aside from parallel tasks.

The general sequence of events is:

- Verify that all required upgrades have arrived complete and site is ready to shut down.
- Shut site down.
- Remove obsolete equipment and cables.

Aside: Removing and installing cables is more efficient and accurate when the rooms are un-cluttered. It is strongly recommended that this be done after equipment is removed and before new equipment is positioned. This house cleaning will make the installation tasks much easier.

- Install new equipment.
- Route new cables
- All discarded material (i.e. removed/discarded hardware, packing material) must be returned in accordance with GE recycling policy
- Power up
- Functional Checks
- System calibrations
- System performance checks
- Complete final tasks and return site to customer.

1-8 OCTANE BILL OF MATERIALS

The following parts lists are specific to the Octane computer upgrade.

CATALOG M1033K REV 1 INDIGO TO OCTANE UPGRADE KIT

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
1	2207030-15	N	COLLECTOR	1	INDY TO OCTANE UPGRADE
2	2204796	N	ASSEMBLY	1	CENTRAL DATA SERIAL EXPANSION & MTG
3A	2154392-2	N	COLLECTOR	1	OPERATOR WORKSTATION CABLES/MRLX2
3B	2154392-3	N	COLLECTOR	ALT	OPERATOR WORKSTATION CABLES/MRLX2
4	2260859	N	DIRECTION	1	SGI UPGRADE INDIGO TO OCTANE
5	2235744-2	1	BIT3 CRD SET	1	VMEBUS ADAPTER CARD
6	2241052-2	2	FIBER OPTIC	1	(836) OW1A15-BIT3 TO MR2A11A30-BIT3
7	2198993	2	CABLE	1	(818) MR2-A11-J31 TO OW1-A16-6,7,8
8	2269111	2	SCREW	4	SCREW-#10 (.19) X 1 INCH ROUND HEAD

INDIGO TO OCTANE UPGRADE 2207030-15 REV 0 (PART 1 OF M1033K)

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
1A	2189500-3	N	ASSEMBLY	1	LX3 SYS COMP ASM W/ 1/100 BASE-T HUB
1B	2189500-2	N	ASSEMBLY	ALT	LX3 SYSTEM COMPUTER ASM
2	2156158-28	1	COLLECTOR	1	LX 9.0 M4 S/W COLLECTOR
5	2186360	N	DIRECTION	1	OW CABINET UNLOADING INSTRUCTIONS
6	46-302200P11	N	RATING PLATE	1	MODEL NUMBER ONLY, NO SERIAL NUMBER
7A	2198999-5	N	ASSEMBLY	1	LX3 OCTANE2 V10 R12KS COMP 360MHZ
7B	2198999-4	N	ASSEMBLY	ALT	ENCORE LX3 OCTANE R12K COMP 270MHZ

LX3 SYSTEM COMPUTER ASM 2189500-3 REV 4 (PART 1A OF 2207030-15)

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
2	2189501-3	N	ASSEMBLY	1	LX2 DUPLEX ASM W/ 10/100 BASE-T HUB
3A	2284297	1	PC COMPUTER	1	MR OP WORKSPACE PC W/-2 BOOT FLOPPY
3B	2192746-3	1	PC WORKSTATI	ALT	LX3 BASE SYSTEM W/2192746-4 S/W
4	2199817-13	N	BOX	1	DUPLEX SCSI EXPANSION BOX W/STAR MOD
5	2270639	N	HUB	1	8 PORT 10/100 BASE T NETWORK HUB W/
7	2301294	N	PART	1	SHIELDED RJ45 BULKHEAD FEEDTHRU
8	2153396	N	PART	2	72 CM LX SHIELD LAN CABLE
9	2148292	N	PART	1	LIGHTNING PC SECURITY KEY CABLE
10	46-269601P5	1	TERMINATOR	1	"SLICK" CIRCUIT ACTIVE TERMINATOR
11	2172316	N	WARNING LABE	1	DUPLEX POWER DISCONNECT WARNING LABL
12	46-328404P1	N	CONFIG.LABEL	1	CONFIGURATION HISTORY LABEL
13	46-302200P4	N	RATING PLATE	1	SERIAL#, MODEL#, VOLTS AMPS HZ, ETC
14	2142857	N	ETL LABEL	1	ETL LABEL
15	2165323	N	LABEL	1	LABEL
16	2160390	N	WARNING LABE	1	DUPLEX POWER WARNING LABEL
17	46-208758P3	N	TY-RAP	3	7.31" X .184" SELF-LOCKING CABLE TIE
18	46-208910P17	N	SCREW	6	SCREW
19	2166263	N	SCREW	2	PHILIPS MACHINE SCREW, PAN HD
20	46-208747P2	N	VIBRATION IS	3	CABLE TIE MOUNT ADHESIVE BACKED
21	46-208758P2	N	TY-RAP	4	5.5" X .140" SELF-LOCKING CABLE TIE.
22	2117390	N	LABELS	1	LABEL, CE MARKING COMPLIANCE
23	2132620-2	N	LABELS	1	CE MARKING LABEL CLASS A GROUP 2

1-8 OCTANE BILL OF MATERIALS (Cont'd)**LX3 OCTANE2 COMPUTER 360 MHZ 2198999-5 REV 0****(PART 7A OF 2207030-15)**

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
2	2169940-60	N	COMPUTER	1	MR LX BASE OCTANE II BUY LEVEL
5	2169940-7	1	COMPUTER	1	OPTION PCI CARD CAGE
7	2169940-9	1	COMPUTER	1	OPTION PCI SCSI CARD
8	2235744-3	1	BIT3 CRD SET	1	PCI ADAPTER CARD
9	46-302200P10	N	RATING PLATE	1	1.72 X 1.10, MODEL#, SN, DATE, DESC.
10	46-328404P1	N	CONFIG.LABEL	1	CONFIGURATION HISTORY LABEL
12	46-208758P2	N	TY-RAP	1	5.5" X .140" SELF-LOCKING CABLE TIE.
13	2245994	N	BRACKET	1	ENCORE FIBER OPTIC STRAIN RELIEF
14	46-208560P112	N	SCREW	2	SCREW, BHMS 4-40 X 3/8 SS

DATA SERIAL EXPANSION BOX 2204796 REV 1**(PART 2 OF M1033K)**

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
2	2189510	1	EXPANSION BO	1	SCSI 8 PORT SERIAL EXPANSION BOX
3	2198593	2	BRACKET	1	SERIAL EXPANSION BOX BRACKET
4	46-170015P48	N	SCREW	4	SCREW
5	46-269601P6	1	TERMINATOR	1	HIGH DENSITY 50-PIN SLICK TERMINATOR

OPERATOR WORKSTATION CABLES 2154392-3 REV 2**(PART 3 OF M1033K)**

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
2	2141980-2	2	CABLE	1	(810) OW1A1A2J8 TO OW1A15 KBD, MOUSE
4	2196990	2	CABLE	1	(814) OW1A1A2J9 TO OW1A2A1 KBD, MOUSE
6	46-328000G958	2	ASSEMBLY	1	(817) OW1A1A2J17 TO OW1A2A5 POWER I/O
7	2188746	2	CABLE	1	(812) OW1A15 TO MODEM LX2 MODEM CBL
8	2168505	2	CABLE	1	CERD SPIKE NOISE TEST CABLE
9	2188747	2	CABLE	1	(813) OW1A2A1COM2 TO OW1A15PORT2
10	2153396-3	2	PART	1	(811) OW1A2A4PT2 HUB TO OW1A15ENET LAN
11	2198995	2	CABLE	1	(819) OW1A15 TO OW1A2A7 LX2 SCSI
12	2198983	2	CABLE	1	(815) OW1A16 TO OW1A15 SCSI

SECTION 2 – POWER OFF & ON PROCEDURES



**FATAL ELECTRIC SHOCK HAZARD!! LETHAL VOLTAGES ARE PRESENT
WITHIN THE PDU EVEN IF ALL PDU BREAKERS ARE OFF.**

2-1 OPERATOR WORKSPACE AND SYSTEM CABINET POWER SHUTDOWN

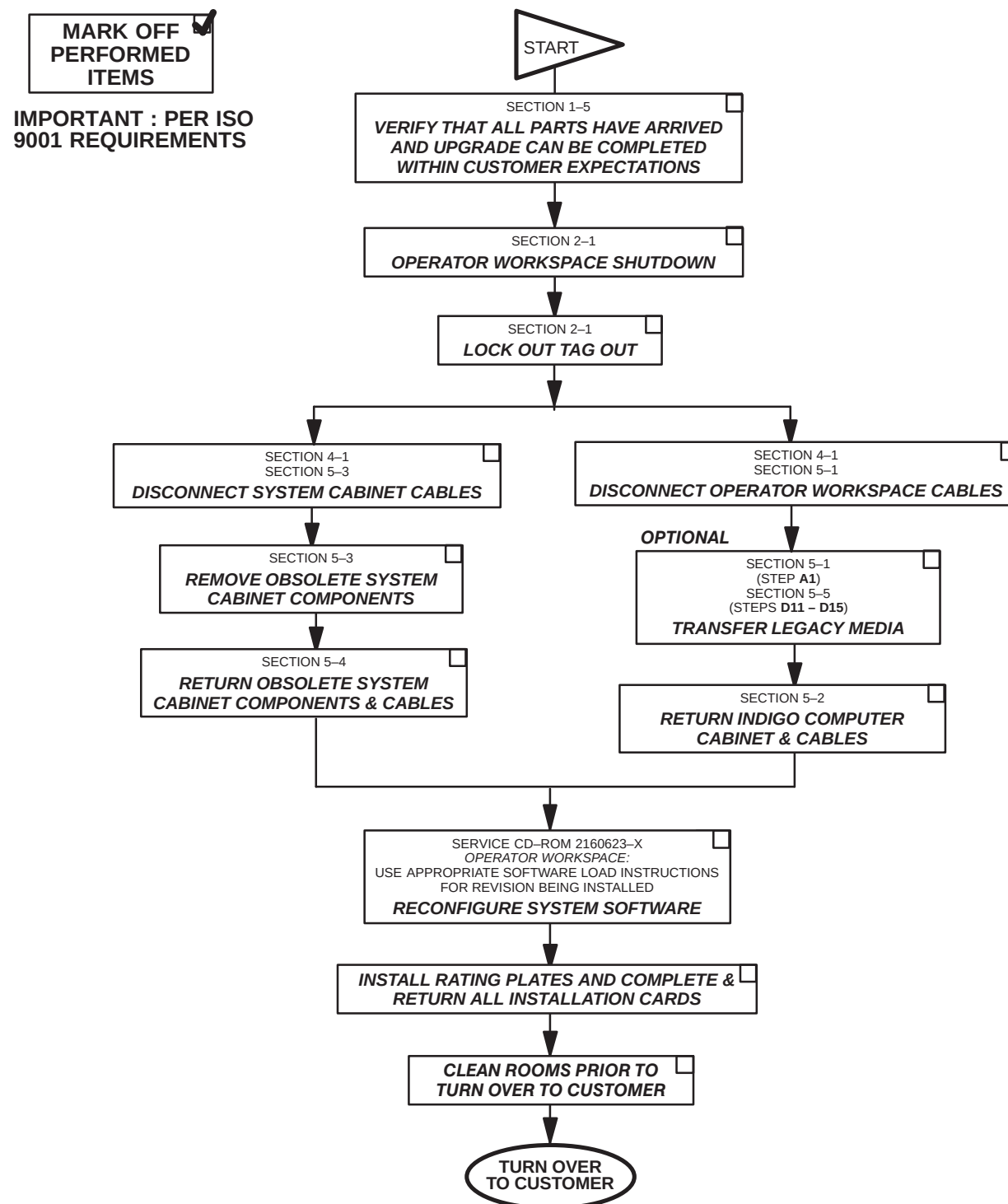
1. Power down monitors.
2. Power down PC.
3. Power down Indigo computer.
4. Power down System cabinet.
5. At the PDU perform lock out and tag out procedures for Operator Workspace Host and PC circuit breakers and System Cabinet circuit breaker.
6. Verify absence of power by attempting to re-energize the PC or Indigo computer.

2-2 OPERATOR WORKSPACE AND SYSTEM CABINET POWER START UP

1. At the PDU remove lock out tags and turn on Operator Workspace Host and PC circuit breakers and System cabinet circuit breaker.
2. Power up System cabinet.
3. Power up Octane computer.
4. Power up PC.
5. Power up monitors.

SECTION 3 – UPGRADE PROCESS

3-1 UPGRADE FLOWCHART

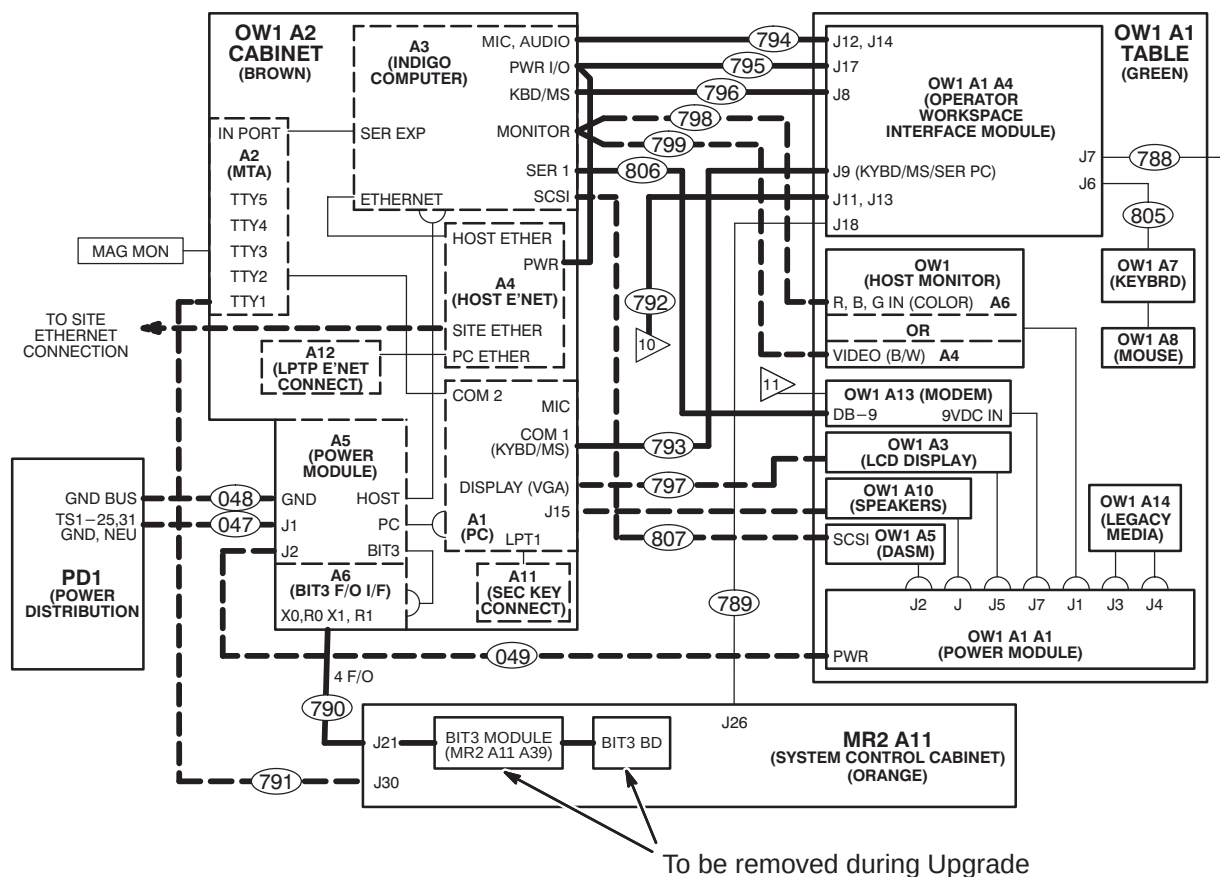


SECTION 4 – CABLES

4-1 INDIGO CABLE REMOVAL FOR UPGRADING TO OCTANE COMPUTER

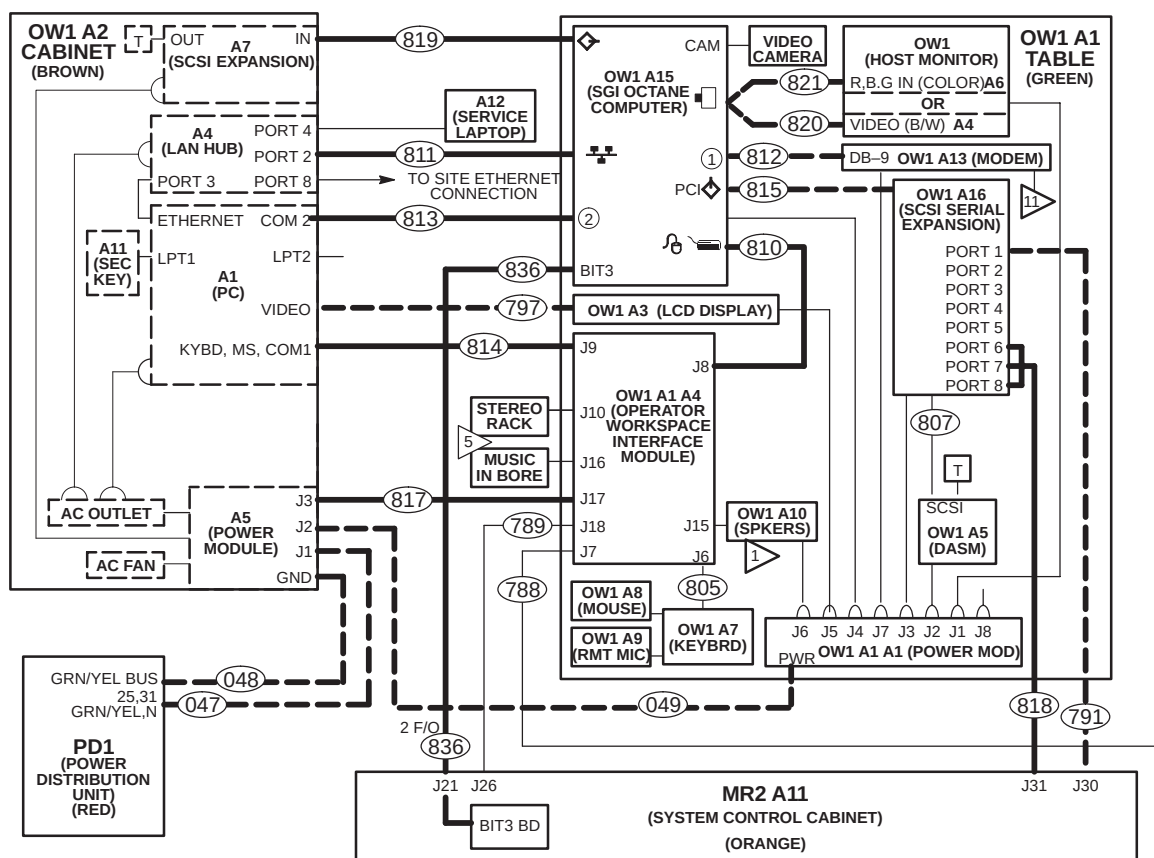
DO NOT CUT CABLES

-  Cables to be disconnected, removed, and returned intact to GEMS Renewable Resources.
 Cables to be disconnected at one end and saved for reconnection to Octane Computer.



4-2 CABLES TO BE INSTALLED AND RECONNECTED FOR NEW OCTANE COMPUTER

- (XXX)— New cables to be connected for Octane Computer upgrade.
- (XXX)— Existing cables to be reconnected for Octane Computer upgrade.



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SECTION 5 – UPGRADE PROCEDURES

5-1 INDIGO COMPUTER DEINSTALLATION PROCEDURE

□ A1 – PREPARE INDIGO COMPUTER FOR DISCONNECTION OF CABLES

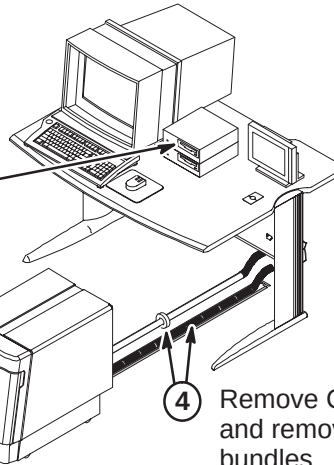
– DO NOT CUT CABLES –

OBSOLETE CABLES ARE TO BE DISCONNECTED AND RETURNED INTACT.

- ① If present, disconnect Pioneer MOD Legacy Media and/or DAT Legacy Media on table. Remove Media Drives from housing. These will be transferred to new Octane Computer Cabinet.

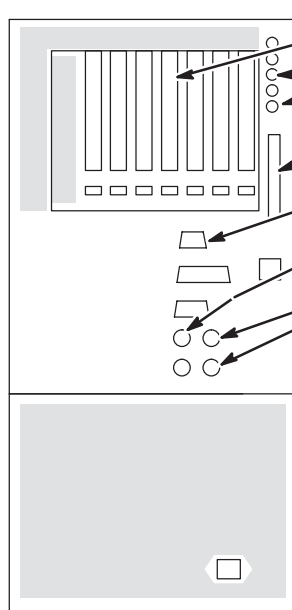
- ② Move Operator Workspace Cabinet (2136000) away from front of table.

- ③ Remove covers.

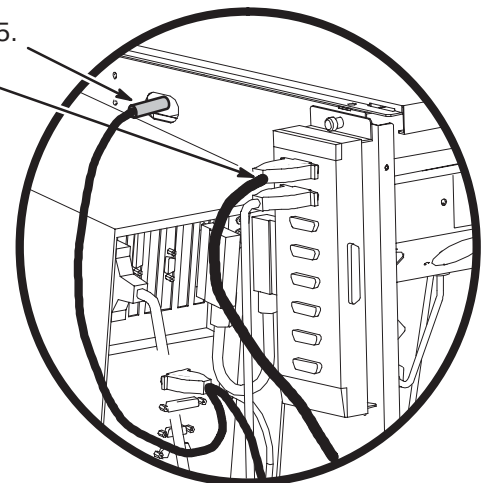
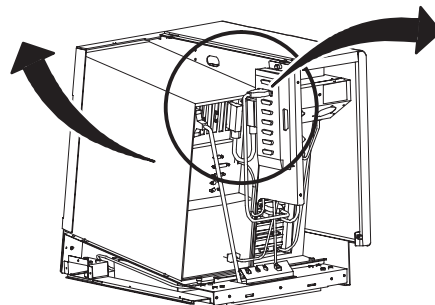


- ④ Remove Clamps, unwrap Cable Covers, and remove ty-wraps around both cable bundles. Save and re-use for Upgrade.

□ A2 – DISCONNECT THE CABLES ON SGI SIDE OF CABINET



- ① Disconnect and save Cable Run 798 (or 799). It will be reused in Upgrade.
- ② Disconnect and return Cable Run 794.
- ③ Disconnect and save for Upgrade SCSI Cable Run 807 from DASM.
- ④ Disconnect and return Power I/O Cable Run 795.
- ⑤ Disconnect and return Cable Run 806 from Modem.
- ⑥ Disconnect and return Mouse/Keyboard Cable Run 796.
- ⑦ Disconnect and save Run 791. Will be reused for upgrade.

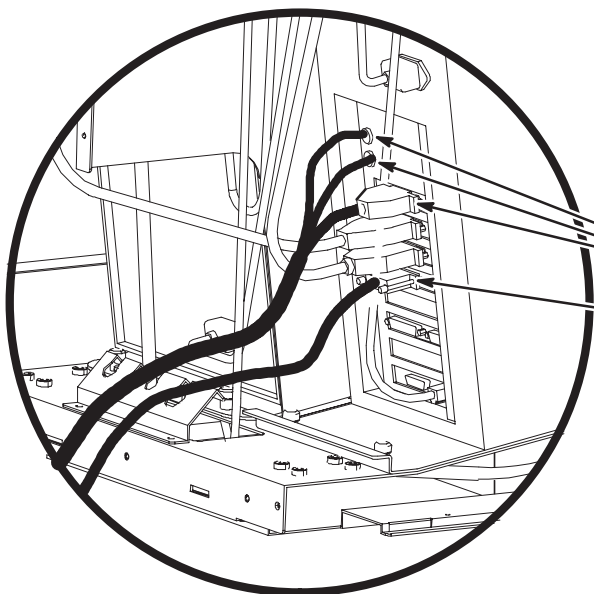


5-1 INDIGO COMPUTER DEINSTALLATION PROCEDURE (CONT'D)

□ A3 – DISCONNECT THE CABLES ON PC MODULE SIDE OF CABINET**– DO NOT CUT CABLES –**

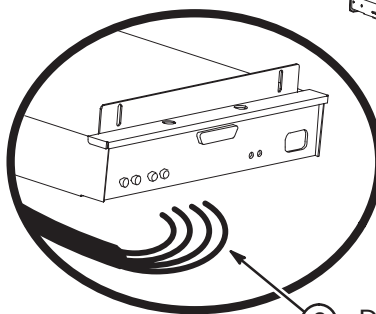
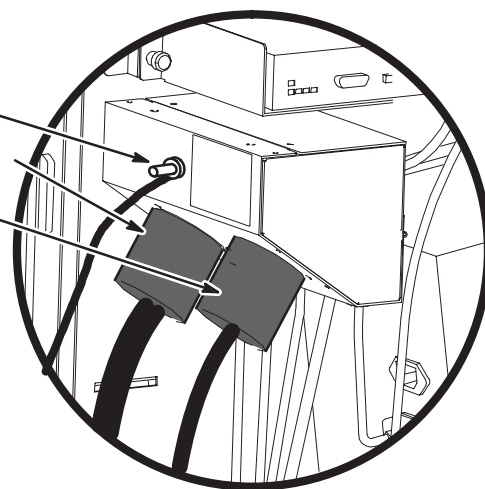
OBSOLETE CABLES ARE TO BE DISCONNECTED
AND RETURNED INTACT.

- ① Disconnect Run 048 and save for reconnection in Upgrade.
- ② Disconnect Run 047 Power Cable from PDU and save for reuse.
- ③ Disconnect Run 050 Power Cable from Operator Workspace Power Module for reuse in Upgrade.



④ Disconnect and return Run 793.

⑤ Disconnect and return Run 797.



⑥ Disconnect and return Run 790, Fiber Optic cables.

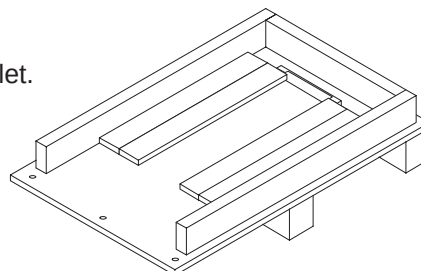
⑦ Replace covers and prepare cabinet for return shipment.
Refer to next section for return procedures.

5-2 INDIGO COMPUTER RETURN PROCEDURE

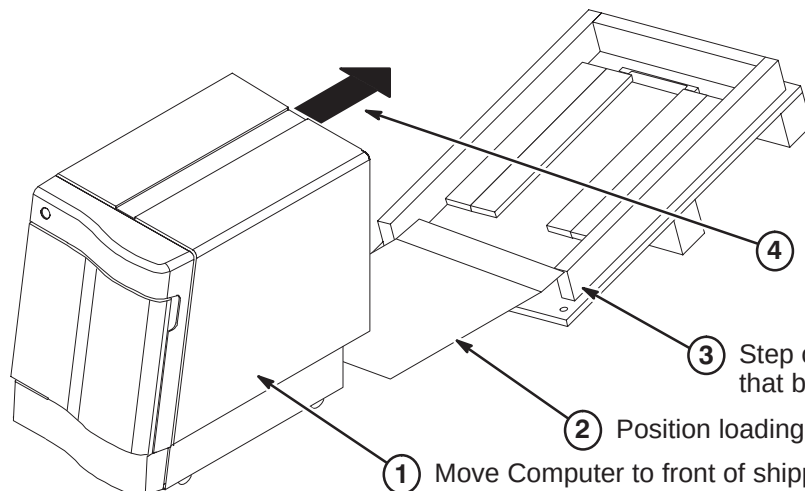
Confirm that all cables are disconnected from Computer. Perform the following procedure for return of Indigo Computer reusing the shipping pallet that the new Operator Workspace Cabinet arrived in.

□ B1 – PREPARING SHIPPING PALLET

- 1 Locate the Operator Workspace Cabinet shipping pallet.



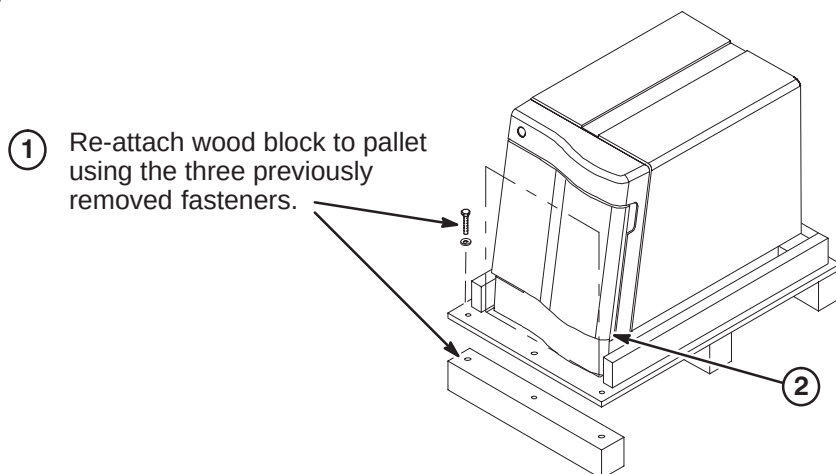
□ B2 – LOADING COMPUTER ONTO SHIPPING PALLET



Note:
Pallet will return to horizontal position as Computer Cabinet is rolled toward back of pallet.

- 4 With another person, carefully roll Computer onto shipping pallet so it will rest on the horizontal foam pieces.
- 3 Step on front corner of shipping pallet so that bottom board of pallet touches floor.
- 2 Position loading board at front of pallet.
- 1 Move Computer to front of shipping pallet.

□ B3 – REASSEMBLE SHIPPING PALLET



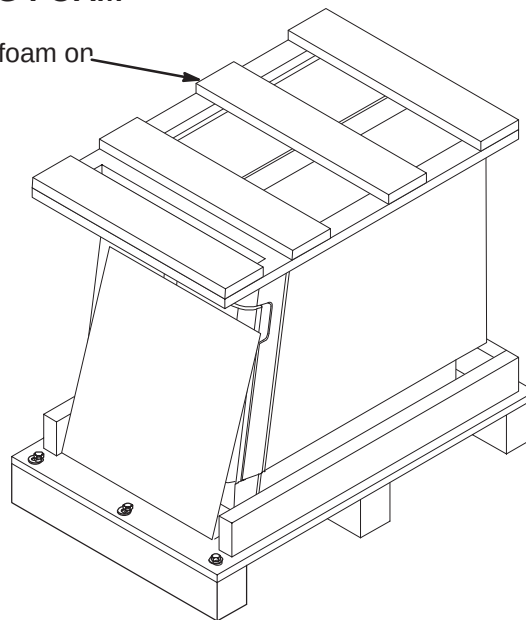
- 1 Re-attach wood block to pallet using the three previously removed fasteners.

- 2 Place 2 x 4 wood block against front right side of Computer and attach to pallet with nails provided.

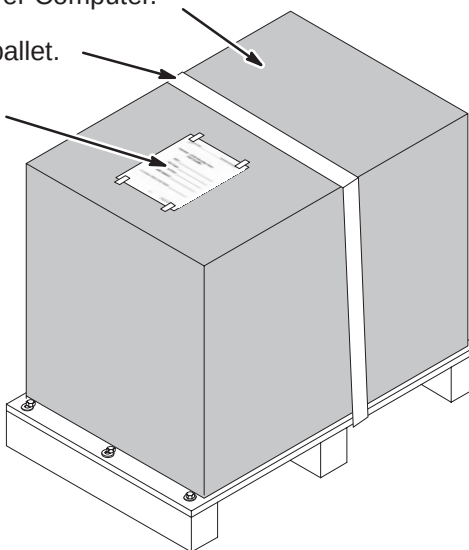
5-2 INDIGO COMPUTER RETURN PROCEDURE (Cont'd)

□ B4 – INSTALLING COMPUTER TOP SHIPPING FOAM

- ① Install the Operator Workspace Cabinet shipping foam on top of Computer.

**□ B5 – FINAL PREPERATION BEFORE SHIPPING**

- ① Place Operator Workspace shipping box over Computer.
- ② Secure to shipping pallet.
- ③ Fill out *GEMS Recycling Form* on next page and attach to top of shipping box.



Return Computer to GE Medical Systems with other equipment that has been replaced during this upgrade.

Attention: GEMS GoldSeal

FDO: _____

Site Name: _____

Removal Date: _____

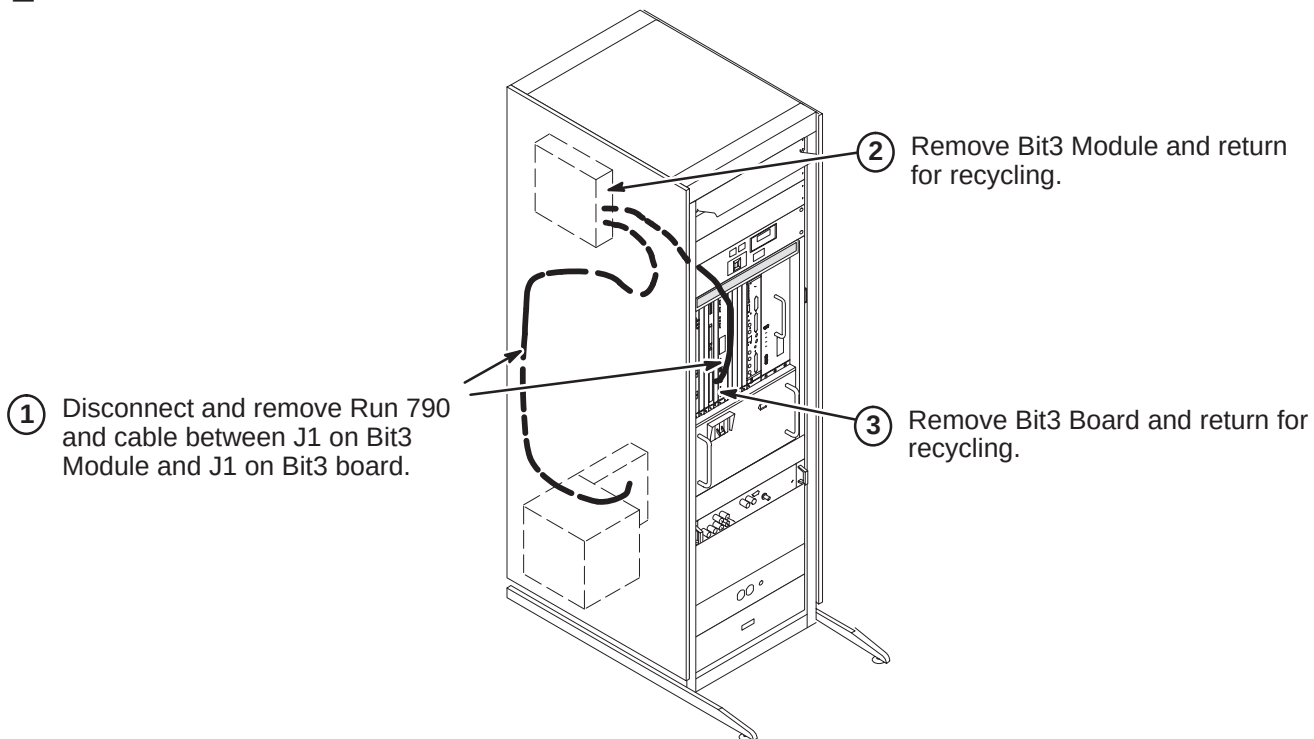
Field Engineer: _____

Indigo Computer Cabinet Serial Number: _____

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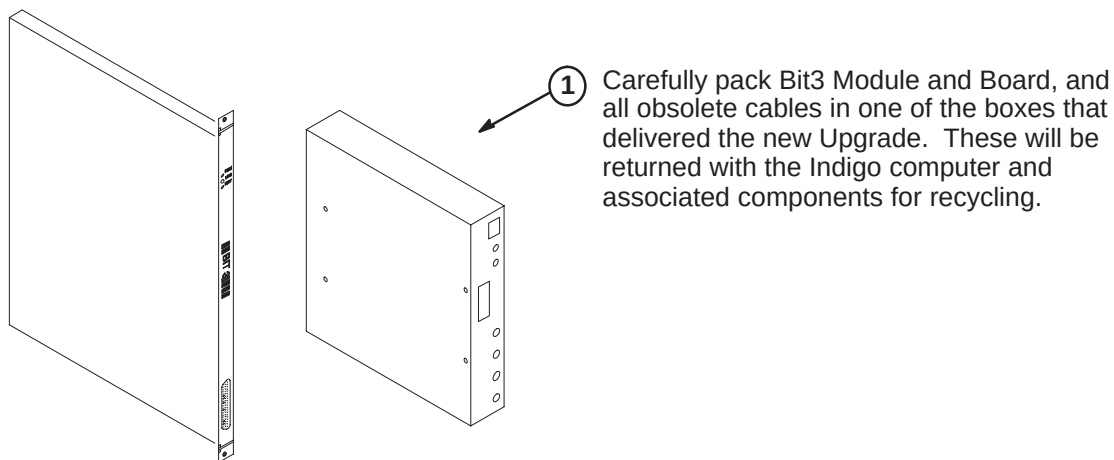
5-3 SYSTEM CABINET DEINSTALLATION

□ C1 – REMOVE OBSOLETE COMPONENTS FROM SYSTEM CABINET



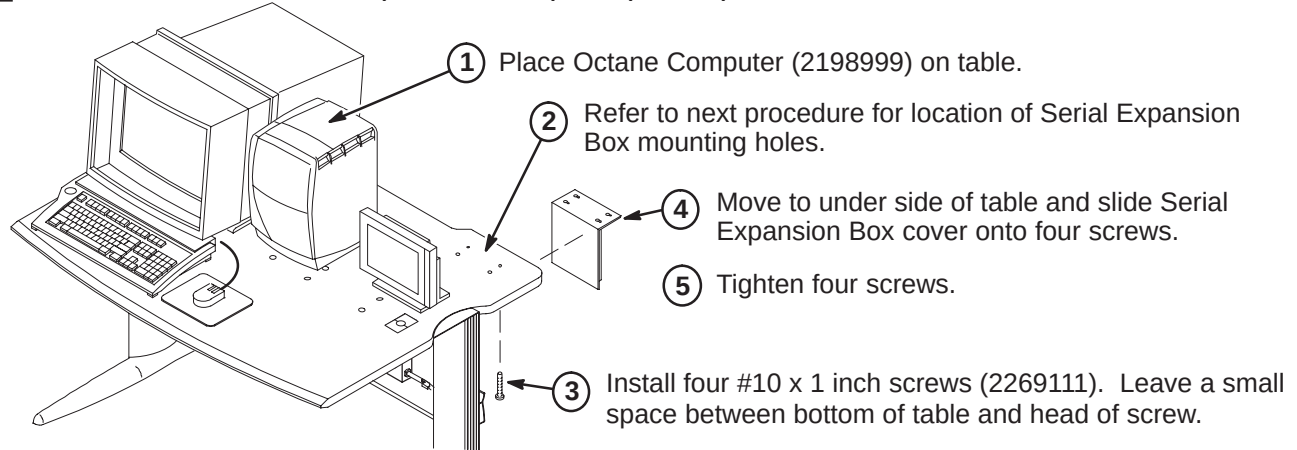
5-4 SYSTEM CABINET COMPONENT RETURN PROCEDURE

□ C1 – RETURN OBSOLETE COMPONENTS & CABLES FROM SYSTEM CABINET

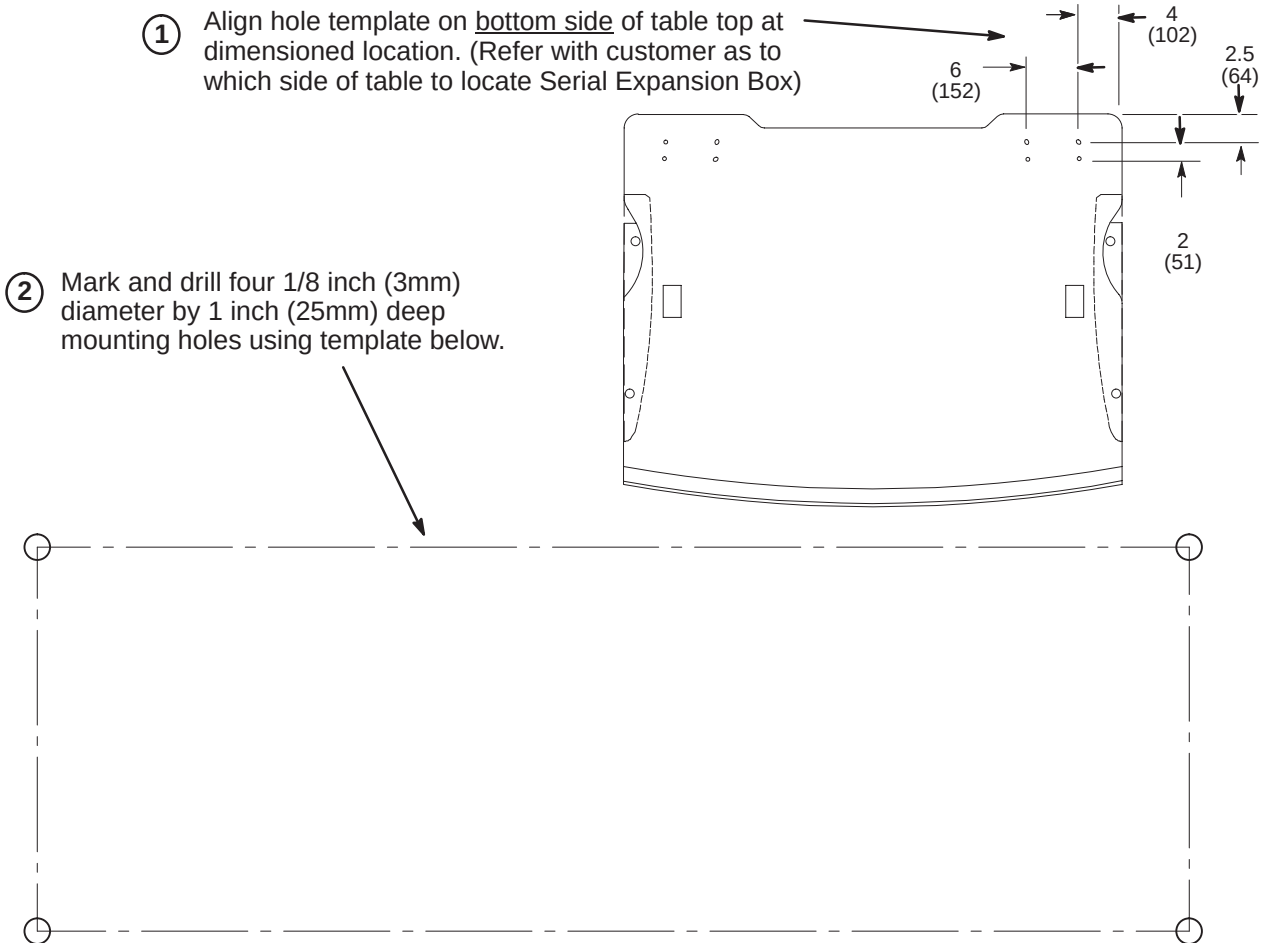


5-5 OCTANE COMPUTER INSTALLATION

□ D1 – INSTALL MOUSE, MONITOR, DAT, MOD, AND LCD DISPLAY ON TABLE



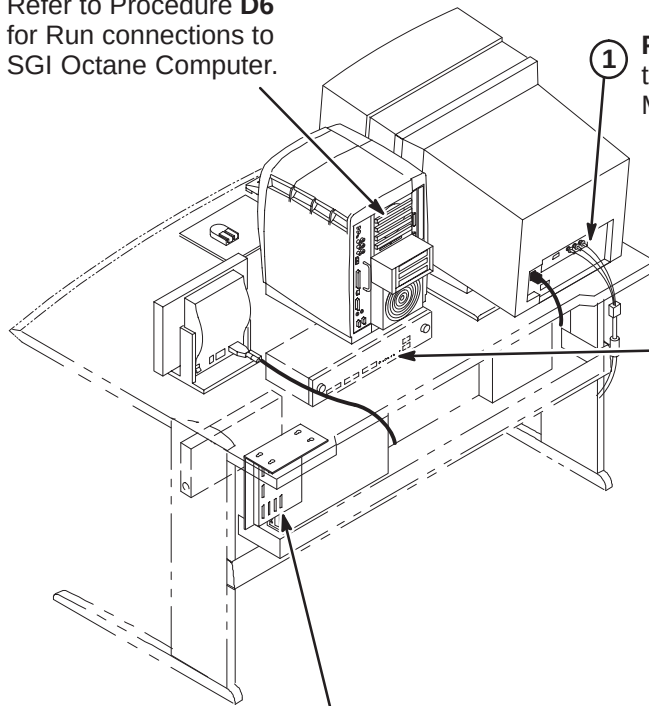
□ D2 – SERIAL EXPANSION BOX MOUNTING LOCATION



5-5 OCTANE COMPUTER INSTALLATION (CONT'D)

D3 – CONNECT TABLE COMPONENT CABLES

Refer to Procedure **D6** for Run connections to SGI Octane Computer.



Refer to Procedure **D4** for Run connections and DIP switch settings.

1 For Color Monitor: Connect Run 798/821 (2150533-2) to R IN, B IN, G IN on Monitor and route/connect to MONITOR on the Octane Computer module.

For Black & White Monitor: Connect Run 799/820 (2150533) to VIDEO on Monitor and route/connect to DISPLAY on the Octane Computer module.

2 Connect the following cables from the OW I/F Module (OW A1 A4) for later connection to Operator Workspace Cabinet:

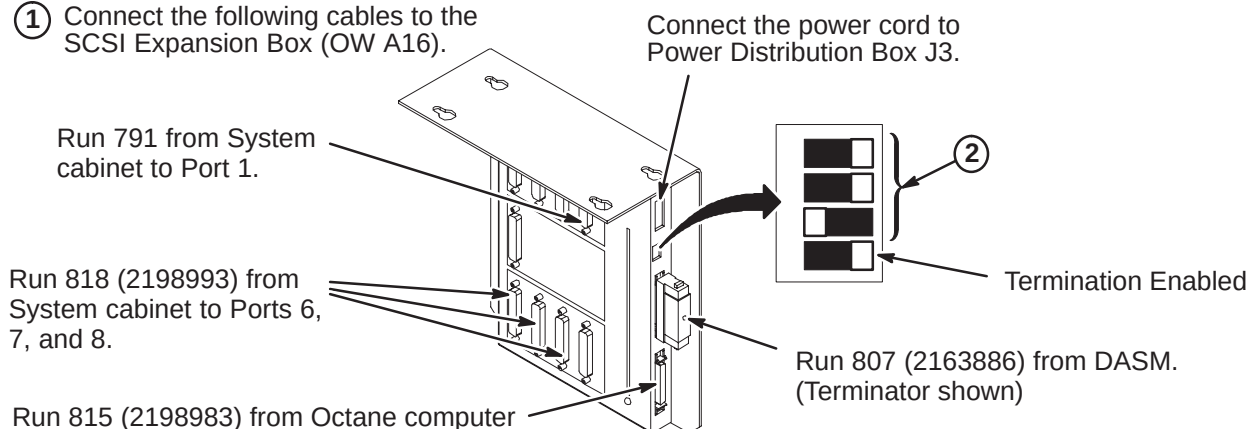
RUN #	CONNECTION	PART #
805	J6	2114561-38
810	J8	2141980-2
814	J9	2196990
817	J17	46-328000G958
	J10	Stereo Cable (option)
	J15	Speaker Cable (option)
	J16	Music in Bore (option)

Connect the following cables to OW I/F Module. These cables will be routed to the System Cabinet and Penetration Panel:

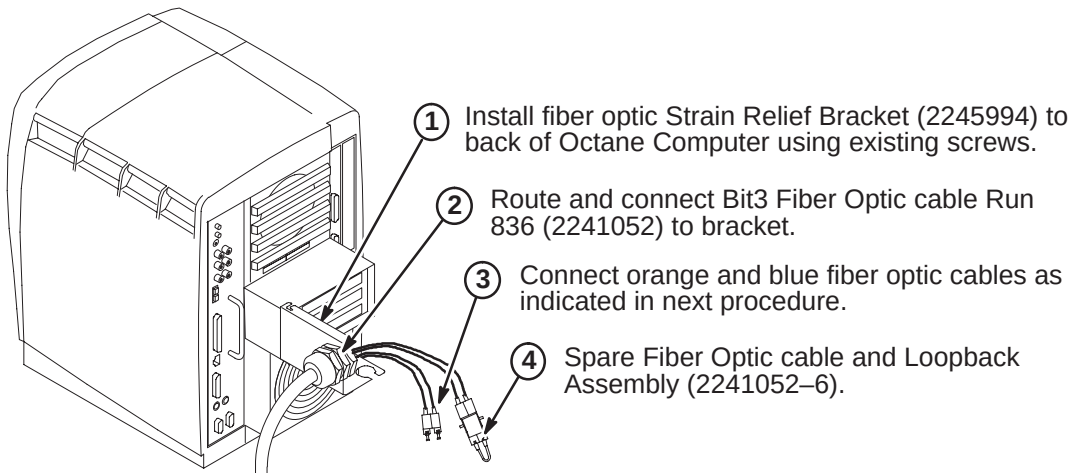
RUN #	CONNECTION	PART #
789	J18	46-317068G938
788	J7	46-328000G937

D4 – SERIAL EXPANSION BOX SET UP

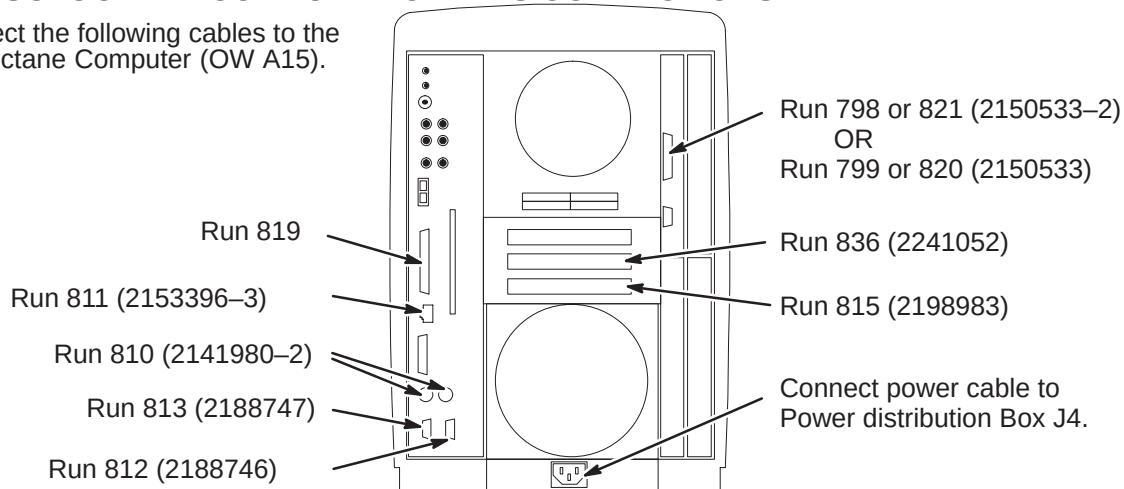
1 Connect the following cables to the SCSI Expansion Box (OW A16).



5-5 OCTANE COMPUTER INSTALLATION (CONT'D)

☐ **D5 – SGI OCTANE RUN 836 INSTALLATION**

☐ **D6 – SGI OCTANE COMPUTER CABLES CONNECTIONS**

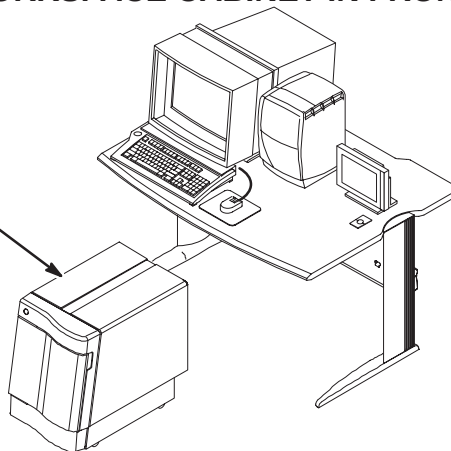
- ① Connect the following cables to the SGI Octane Computer (OW A15).



5-5 OCTANE COMPUTER INSTALLATION (CONT'D)

D7 – LOCATE OPERATOR WORKSPACE CABINET IN FRONT OF TABLE

- ① Position new Octane Operator Workspace Cabinet in front of table.

**D8 – REMOVE COMPUTER CABINET REAR, FRONT, AND SIDE COVERS**

- ④ Unfasten screw and slide Right Side Cover backwards to remove.

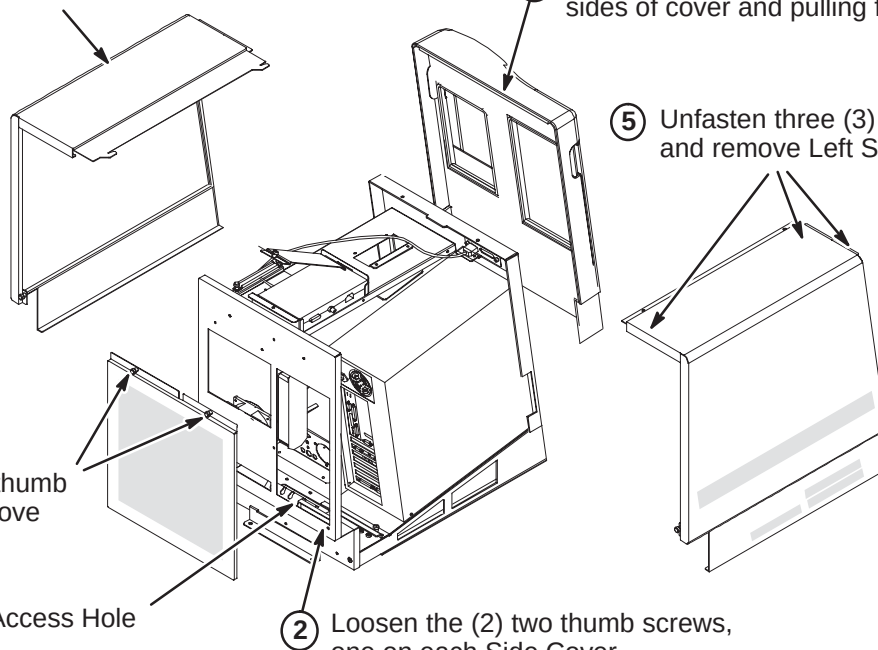
- ③ Remove front cover by grasping both sides of cover and pulling forward.

- ⑤ Unfasten three (3) screws and remove Left Side Cove.

- ① Loosen (2) two thumb screws and remove Rear Cover.

Cable Access Hole

- ② Loosen the (2) two thumb screws, one on each Side Cover.

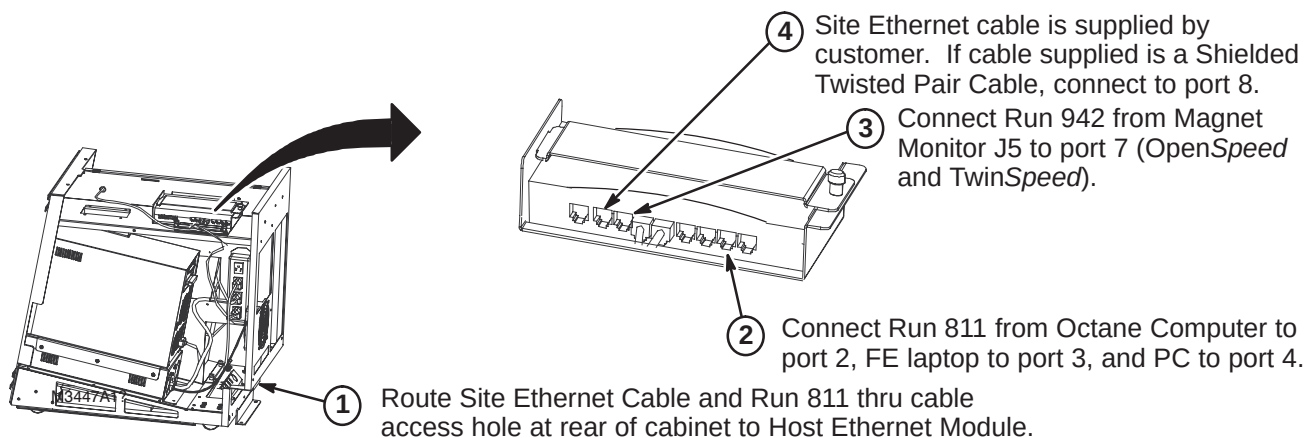


5-5 OCTANE COMPUTER INSTALLATION (CONT'D)

E4 – CONNECT ETHERNET CABLES

Note:

Site Ethernet cable for InSite connection is supplied by customer. For Europe, connection to ISDN router is required. Refer to catalog required for country or region for specific equipment to be installed.



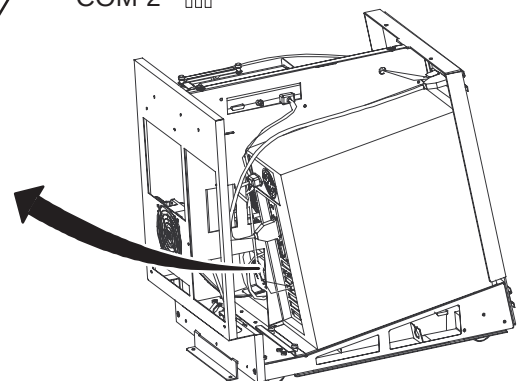
E5 – ROUTE/CONNECT RUNS 792, 797, 813, AND 814 TO PC MODULE

Note: Route cables thru access hole at rear of cabinet.

- ① Route Run 814 and connect as marked to:
- KEYBOARD 
 - MOUSE 
 - COM1 

- ② Route Run 797 and connect as marked to:
- VIDEO 

- ③ Route Run 813 and connect as marked to:
- COM 2 

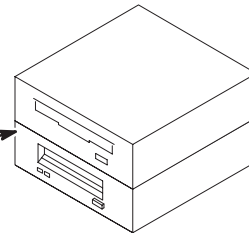


5-5 OCTANE COMPUTER INSTALLATION (CONT'D)

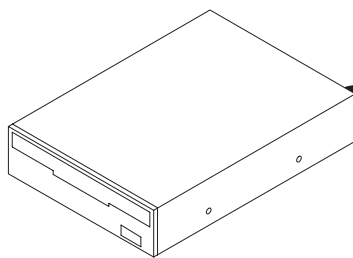
□ D11 – REMOVE DAT AND MOD DRIVES (LEGACY MEDIA) FROM HOUSINGS

Note: Complete procedures **D11** thru **D15** only if Legacy Media is present on existing system and is being transferred to new Octane Computer cabinet.

- ① Remove Media Drives from housing for transfer to new Octane Computer Cabinet.

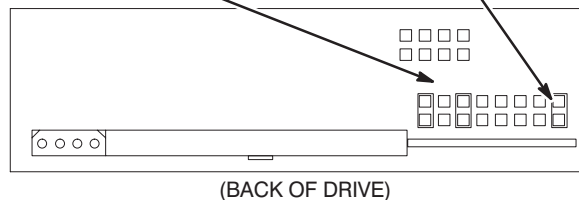


□ D12 – MOD DRIVE SETTINGS

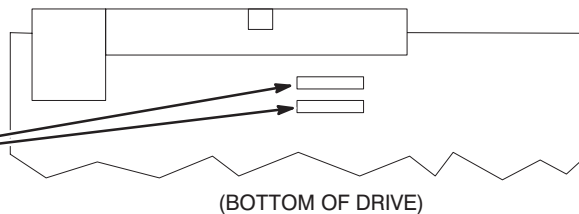


- ② For Octane, SCSI ID = 5

- ③ Make sure jumper is installed on pins at position 8 for SCSI Optical Disk.



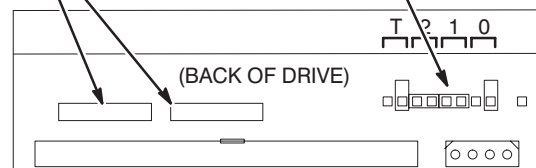
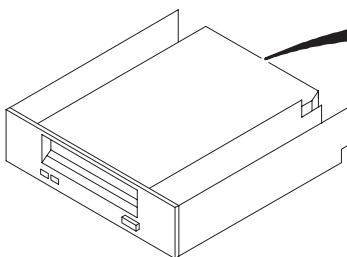
- ① Confirm that SCSI Bus Resistor Terminator Packs are not installed. If present – remove them.



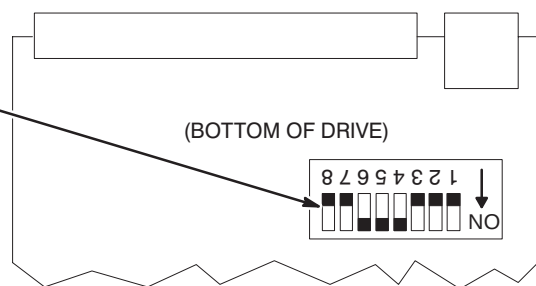
□ D13 – DAT DRIVE SETTINGS

- ① Confirm that SCSI Bus Resistor Terminator Packs are not installed. If present – remove them.

- ② For Octane, set SCSI ID = 6.

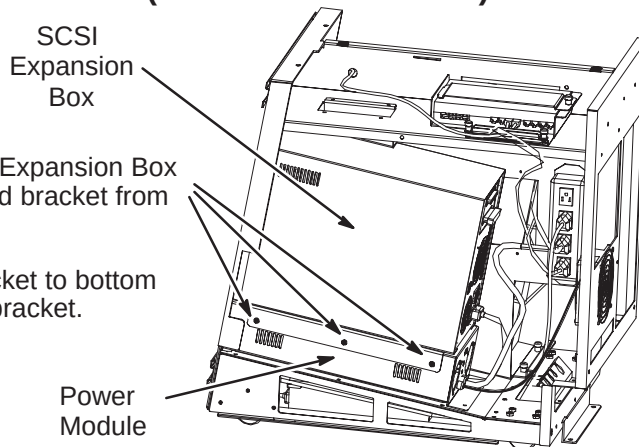


- ③ Set the switches on the bottom of the drive as shown. This sets the drive to NOT perform "data disconnects" when there is no data available. This is required because of the odd byte count record packing done on GEMS tapes and SGI's inability to handle odd byte "save pointer" SCSI requests.

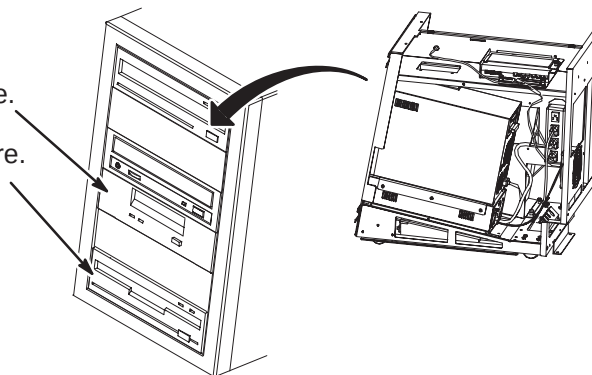


5-5 OCTANE COMPUTER INSTALLATION (CONT'D)**□ D14 – REMOVE SCSI EXPANSION BOX (FOR LEGACY MEDIA)**

- ① Unfasten three (3) screws from SCSI Expansion Box mounting bracket and remove Box and bracket from power module.
- ② Unfasten six (6) screws that hold bracket to bottom of SCSI Expansion Box and remove bracket.
- ③ Remove SCSI Expansion Box cover.

**□ D15 – INSTALL DAT & MOD DRIVES (LEGACY MEDIA) IN SCSI EXPANSION BOX**

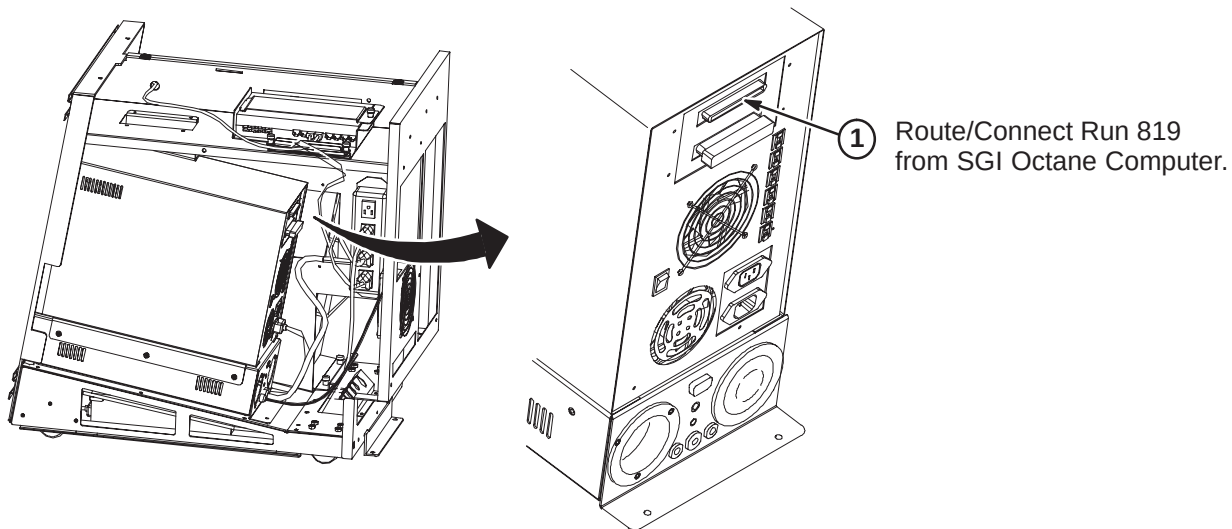
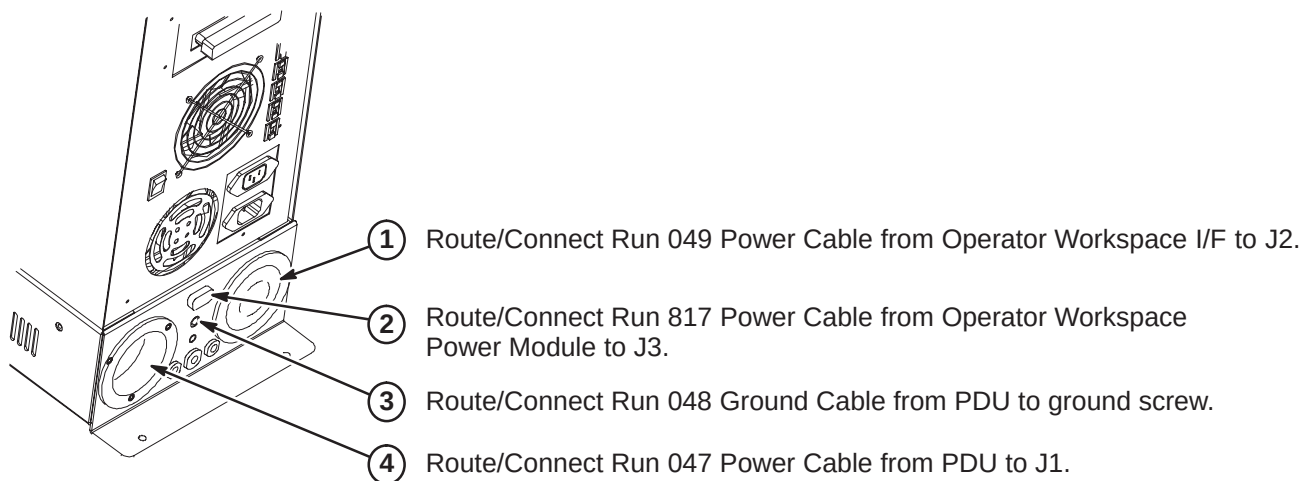
- ① Install DAT Drive in slot 4 with existing hardware.
- ② Install MOD Drive in slot 6 with existing hardware.
- ③ Connect power cords to DAT and MOD Drives.
- ④ Connect ribbon cable to DAT and MOD Drives.
- ⑤ Install SCSI Expansion Box cover. Reattach bracket to bottom of Box. Reattach bracket and Box assembly to power module.



5-5 OCTANE COMPUTER INSTALLATION (CONT'D)

□ D16 – ROUTE/CONNECT RUN 819 TO SCSI EXPANSION BOX

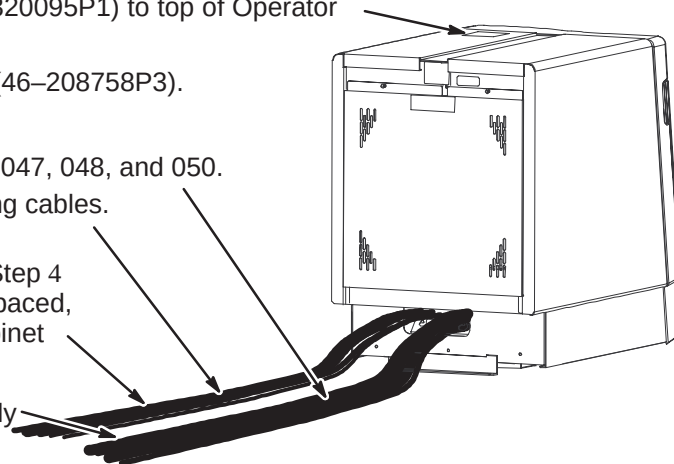
Note: Route all cables from SGI Octane Computer thru cable access hole at rear of cabinet.

**□ D17 – ROUTE/CONNECT RUNS 047, 048, 049, AND 817 TO POWER MODULE**

5-5 OCTANE COMPUTER INSTALLATION (CONT'D)

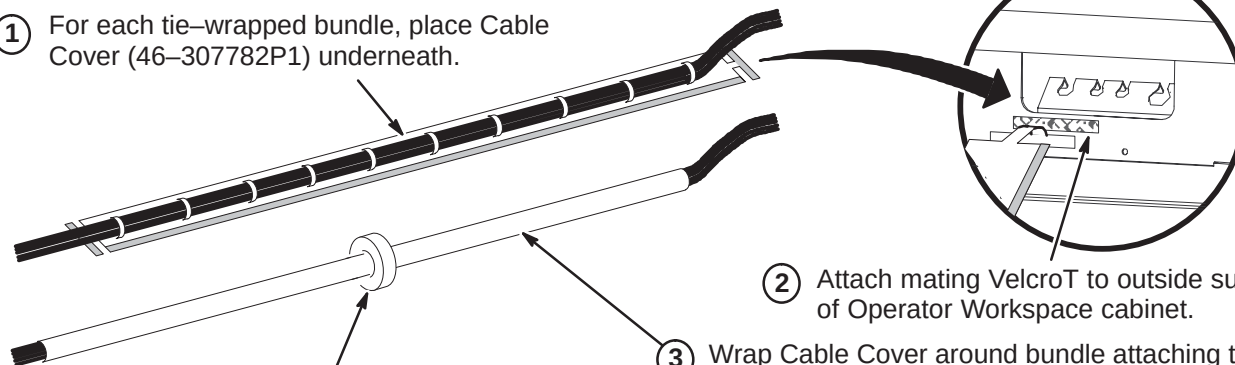
□ D18 – ALIGN OPERATOR WORKSPACE CABINET CABLES & INSITE MAGNET

- ① Install Operator Workspace Cabinet right and left Side covers, Front cover and Rear cover.
- ② Transfer the "Computer" cabinet magnet (46-320095P1) to top of Operator Workspace cabinet.
- ③ Locate twenty (20) seven inch long cable ties (46-208758P3).
- ④ Separate cables into two bundles;
The first bundle will include power cable Runs 047, 048, and 050.
The second bundle will include all the remaining cables.
- ⑤ Align the cables of the first bundle created in Step 4 and attach ten (10) of the cable ties, equally spaced, around the cable bundle between the OW Cabinet and the OW Table.
- ⑥ Attach the remaining ten (10) cable ties, equally spaced, around the second cable bundle.



□ D19 – BUNDLE OPERATOR WORKSPACE CABINET CABLES

- ① For each tie-wrapped bundle, place Cable Cover (46-307782P1) underneath.
- ② Attach mating VelcroT to outside surface of Operator Workspace cabinet.
- ③ Wrap Cable Cover around bundle attaching the VelcroT hooks and loops.
- ④ Attach cable bundle Clamp (46-307873P1) to center of Cable Cover.
- ⑤ Carefully roll OW Cabinet into final position. Make sure cable bundles are unobstructed and slide easily. Attach to mating Velcro on cabinet.
- ⑥ Repeat Step 4 as often as necessary.



5-6 SYSTEM CABINET UPGRADE INSTALLATION

E1 – ROUTE/CONNECT FIBER OPTIC RUN 836 TO BIT3 MODULE

Note: On some early versions of the 8.x System cabinet, service cable 2198994 (LX2 Debug, I/F J31 to CERD, IPG, CPU), was not included. If the System Cabinet on this site does not have this cable and you would like to install it, place an order. The system can operate without this cable in the mean time.

